

Bike Station Utilization Analysis

OBJECTIVES

Develop an interactive Power BI Dashboard to analyze bike station availability, occupancy trends, operational status, and station utilization for better decision making and resource optimization.

SCREENSHOTS

DATA CLEANING

Import Data & Transform Data

Navigator

Display Options ▾

Bike Station.xlsx [2]

☒ Table1

☐ Sheet1

Table1

number	name	address
2000	2000 - TEST DSI PLAISIR	
1297	1297- ST FÉRRÉOL DAVSO	ST FERREOL DAVSO - RUE FRANCIS DAVSO
8265	8265- NÉGRESKO - PAULET	NEGRESKO PAULET - FACE AU 42 RUE M
8065	8065-PROMENADE POMPIDOU PALM BEA	PROMENADE POMPIDOU PALM BEACH
1011	01011 - BORNE TEST NANTES 1	Borne test 1
4342	4342 - FLAMMARION MONTRICHET	FLAMMARION MONTRICHET - PLACE LE
6176	6176-CANTINI ROUET	CANTINI ROUET - FACE AU 27 AVENUE
3322	3322 - ARENC CHANTERAC	32 rue de Chanterac 13003 Marseille
1264	1264 - CANEBIÈRE DUGOMMIER	CANEBIERE DUGOMMIER - LA CANEBIE
1191	1191-CANEBIERE-ST FERREOL	CANEBIERE SAINT FERREOL - ANGLE CA
5058	5058 -SAKAKINI - SAINT PIERRE	SAKAKINI SAINT PIERRE - BOULEVARD S
4293	4293- BLANCARDE	BLANCARDE - GARE DE LA BLANCARDE
1245	1245-GAMBETTA - LAFAYETTE	GAMBETTA LAFAYETTE - 6 ALLEES LEON
7153	7153-CORSE ST MAURICE	CORSE SAINT MAURICE - 28 AVENUE DI
6154	6154-PIERRE PUGET BRETEUIL	PIERRE PUGET BRETEUIL - COURS PIERF
1002	1002-HOTEL DE VILLE	HOTEL DE VILLE - Face au 42 QUAI DU F
1224	1224-PLAINE - BIBLIOTHEQUE	PLAINE BIBLIOTHEQUE - 45 RUE DE LA I
1287	1287- STALINGRAD - RÉFORMÉS	STALINGRAD REFORMES - SQUARE STA
500	500 - IT PLAISIR	
5220	5220-ROOSEVELT - ORAN	ROOSEVELT ORAN - 53 COURS FRANKLI
6160	6160-153 PARADIS	153 PARADIS - 153 RUE PARADIS
7026	7026-CATALANS CORNICHE	CATALANS CORNICHE - 69 CORNICHE J.
54000	54000 - TEST IT	

Load

Transform Data

Cancel

Remove Null Values

Query Settings

Table1

APPLIED STEPS

- Source
- Navigation
- Changed Type
- Renamed Columns into Proper Case
- Remove null rows

Query Formula: = Table.SelectRows(#"Renamed Columns into Proper Case", each ([Number] <> null))

Number	Name	Address	Position
2000	2000 - TEST DSI PLAISIR		null
1297	1297- ST FÉRRÉOL DAVSO	ST FERREOL DAVSO - RUE FRANCIS DAVSO ANGLE RUE SAINT FERREOL	43.293485736186526, 5.376
8265	8265- NÉGRESKO - PAULET	NEGRESKO PAULET - FACE AU 42 RUE NEGRESKO	43.269318556371154, 5.38
8065	8065-PROMENADE POMPIDOU PALM ...	PROMENADE POMPIDOU PALM BEACH - 6 PROMENADE GEORGES PO...	43.267272515671856, 5.37
01011	01011 - BORNE TEST NANTES 1	Borne test 1	47.19504, -1.5575
4342	4342 - FLAMMARION MONTRICHET	FLAMMARION MONTRICHET - PLACE LE VERRIER ANGLE BD FLAMMAR...	43.305993453766845, 5.39
6176	6176-CANTINI ROUET	CANTINI ROUET - FACE AU 27 AVENUE JULES CANTINI	43.284775983612235, 5.38
3322	3322 - ARENC CHANTERAC	32 rue de Chanterac 13003 Marseille	43.3123370339008, 5.3687
1264	1264 - CANEBIÈRE DUGOMMIER	CANEBIERE DUGOMMIER - LA CANEBIERE ANGLE BOULEVARD DUGO...	43.29789631329341, 5.3811
1191	1191-CANEBIERE-ST FERREOL	CANEBIERE SAINT FERREOL - ANGLE CANEBIERE SAINT FERREOL	43.29620767061392, 5.376
5058	5058 -SAKAKINI - SAINT PIERRE	SAKAKINI SAINT PIERRE - BOULEVARD SAKAKINI ANGLE RUE SAINT PIE...	43.292800863545835, 5.401
4293	4293- BLANCARDE	BLANCARDE - GARE DE LA BLANCARDE	43.29626340517055, 5.406
1245	1245-GAMBETTA - LAFAYETTE	GAMBETTA LAFAYETTE - 6 ALLEES LEON GAMBETTA	43.29902793826475, 5.381
7153	7153-CORSE ST MAURICE	CORSE SAINT MAURICE - 28 AVENUE DE LA CORSE ANGLE RAMPE SAIN...	43.28946760889984, 5.363
6154	6154-PIERRE PUGET BRETEUIL	PIERRE PUGET BRETEUIL - COURS PIERRE PUGET ANGLE RUE BRETEUIL	43.290329770206334, 5.37

Replace Values

Query Settings

Table1

APPLIED STEPS

- Source
- Navigation
- Changed Type
- Renamed Columns into Proper Case
- Remove null rows
- Replaced Value É into E
- Replaced Value Ô into O
- Replaced Value À into A
- Replaced Value Ê into E

Query Formula: = Table.ReplaceValue(#"Remove null rows", "É", "E", Replacer.ReplaceText, {"Name"})

Replace Values

Replace one value with another in the selected columns.

Value To Find: É

Replace With: E

Advanced options

OK Cancel

Number	Name	Address	Position
2000	2000 - TEST DSI PLAISIR		
1297	1297- ST FÉRRÉOL DAVSO	ST FERREOL DAVSO - RUE FRANCIS DAVSO ANGLE RUE SAINT FERREOL	
8265	8265- NÉGRESKO - PAULET	NEGRESKO PAULET - FACE AU 42 RUE NEGRESKO	
8065	8065-PROMENADE POMPIDOU PALM ...	PROMENADE POMPIDOU PALM BEACH - 6 PROMENADE GEORGES PO...	
01011	01011 - BORNE TEST NANTES 1	Borne test 1	
4342	4342 - FLAMMARION MONTRICHET	FLAMMARION MONTRICHET - PLACE LE VERRIER ANGLE BD FLAMMAR...	
6176	6176-CANTINI ROUET	CANTINI ROUET - FACE AU 27 AVENUE JULES CANTINI	
3322	3322 - ARENC CHANTERAC	32 rue de Chanterac 13003 Marseille	43.3123370339008, 5.3687418043728
1264	1264 - CANEBIÈRE DUGOMMIER	CANEBIERE DUGOMMIER - LA CANEBIERE ANGLE BOULEVARD DUGO...	43.29789631329341, 5.381132215559973
1191	1191-CANEBIERE-ST FERREOL	CANEBIERE SAINT FERREOL - ANGLE CANEBIERE SAINT FERREOL	43.29620767061392, 5.3769306220596
5058	5058 -SAKAKINI - SAINT PIERRE	SAKAKINI SAINT PIERRE - BOULEVARD SAKAKINI ANGLE RUE SAINT PIE...	43.292800863545835, 5.401285286456447
4293	4293- BLANCARDE	BLANCARDE - GARE DE LA BLANCARDE	43.29626340517055, 5.40600080377506
1245	1245-GAMBETTA - LAFAYETTE	GAMBETTA LAFAYETTE - 6 ALLEES LEON GAMBETTA	43.29902793826475, 5.381500046726694
7153	7153-CORSE ST MAURICE	CORSE SAINT MAURICE - 28 AVENUE DE LA CORSE ANGLE RAMPE SAIN...	43.28946760889984, 5.36370990786361
6154	6154-PIERRE PUGET BRETEUIL	PIERRE PUGET BRETEUIL - COURS PIERRE PUGET ANGLE RUE BRETEUIL	43.290329770206334, 5.375548341850035

Remove Unnecessary Numbers & Letters in Name Column

Custom Column

Add a column that is computed from the other columns.

New column name
Custom Name

Custom column formula ⓘ
= Text.Select([Name],{"A".."Z","a".."z"," ","-"}))

Available columns
Number
Name
Address
Position
Banking
Bonus
Status

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK Cancel

Query Settings

Name
Bike Station

All Properties

APPLIED STEPS

- Source
- Navigation
- Changed Type
- Renamed Columns into Pr...
- Remove null rows
- Replaced Value É into E
- Replaced Value Ô into O
- Replaced Value À into A
- Replaced Value È into E
- Added Custom
- Reordered Columns
- Changed Type1
- Capitalized Each Word in ...
- Removed Columns
- Replaced Value
- Trimmed Text
- Removed Columns1

Replace - Value

Replace Values

Replace one value with another in the selected columns.

Value To Find
-

Replace With

Advanced options

OK Cancel

Query Settings

Name
Bike Station

All Properties

APPLIED STEPS

- Source
- Navigation
- Changed Type
- Renamed Columns into Prop...
- Remove null rows
- Replaced Value É into E
- Replaced Value Ô into O
- Replaced Value À into A
- Replaced Value È into E
- Added Custom
- Reordered Columns
- Changed Type1
- Capitalized Each Word in Na...
- Removed Columns
- Replaced Value
- Trimmed Text

Remove Unnecessary Letters in Address Column

Custom Column

Add a column that is computed from the other columns.

New column name: Custom

Custom column formula: `= Text.Select([Address],{"A"."Z","a"."z","0"."9"," ","-","."})`

Available columns: Number, Name, Address, Position, Banking, Bonus, Status

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK Cancel

Query Settings

PROPERTIES

Name: Bike Station

All Properties

APPLIED STEPS

- Remove null rows
- Replaced Value É into E
- Replaced Value Ô into O
- Replaced Value À into A
- Replaced Value Ê into E
- Added Custom
- Reordered Columns
- Changed Type1
- Capitalized Each Word in ...
- Removed Columns
- Replaced Value
- Trimmed Text
- Removed Columns1
- Renamed Columns
- Replaced Value1
- Changed Type Date/Timez...
- Added Custom1

Position Column Split into Latitude & Longitude

Split Column by Delimiter

Specify the delimiter used to split the text column.

Select or enter delimiter: --Custom--

Delimiter: ,

Split at:

- ☒ Left-most delimiter
- ☐ Right-most delimiter
- ☐ Each occurrence of the delimiter

Advanced options

Quote Character: "

☐ Split using special characters

Insert special character

OK Cancel

Query Settings

PROPERTIES

Name: Bike Station

All Properties

APPLIED STEPS

- Replaced Value À into A
- Replaced Value Ê into E
- Added Custom
- Reordered Columns
- Changed Type1
- Capitalized Each Word in ...
- Removed Columns
- Replaced Value
- Trimmed Text
- Removed Columns1
- Renamed Columns
- Replaced Value1
- Changed Type Date/Timez...
- Added Address Column
- Reordered Columns1
- Removed Columns2
- Renamed Columns1

Latitude & Longitude

Query Settings

PROPERTIES

Name

Bike Station

All Properties

APPLIED STEPS

Reordered Columns

Changed Type1

Capitalized Each Word in ...

Removed Columns

Replaced Value

Trimmed Text

Removed Columns1

Renamed Columns

Replaced Value1

Changed Type Date/Timez...

Added Address Column

Reordered Columns1

Removed Columns2

Renamed Columns1

Split Column by Delimiter

Changed Type2

Renamed Columns2

Table: RenameColumns(#"Changed Type2",{"Position.1", "Latitude"}, {"Position.2", "Longitude"})

	1.2 Latitude	1.2 Longitude	Banking	Bonus	Status
	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty
	999 distinct, 998 unique	1000 distinct, 1000 unique	2 distinct, 0 unique	2 distinct, 0 unique	2 distinct, 0 unique
1	49.901646	2.303244	FALSE	FALSE	OPEN
2	59.95363	11.04428	FALSE	FALSE	CLOSED
3	Castelar	43.461719	TRUE	FALSE	CLOSED
4	ssigny	48.7988351	TRUE	FALSE	OPEN
5		45.72463042	FALSE	FALSE	OPEN
6		48.79995838	FALSE	FALSE	OPEN
7		49.43952415	TRUE	FALSE	CLOSED
8		47.742424	TRUE	FALSE	CLOSED
9		55.7065203	TRUE	FALSE	OPEN
10		36.705842	FALSE	FALSE	OPEN
11		46.051367	FALSE	FALSE	OPEN
12	berto Jimnez-Becerril	37.41292355	FALSE	FALSE	OPEN
13	Jordana	39.48001	FALSE	FALSE	OPEN
14	103	49.598211	FALSE	FALSE	OPEN
15	ux Hall Omnisport	50.47536	FALSE	FALSE	OPEN

Applied Steps

PROPERTIES

Name

Bike Station

All Properties

APPLIED STEPS

Renamed Columns into Pr...	
Remove null rows	*
Replaced Value É into E	*
Replaced Value Ô into O	*
Replaced Value À into A	*
Replaced Value È into E	*
Added Custom	*
Reordered Columns	
Changed Type1	
Capitalized Each Word in ...	
Removed Columns	
Replaced Value	*
Trimmed Text	
Removed Columns1	
Renamed Columns	
Replaced Value1	*
Changed Type Date/Timez...	

DATA TRANSFORMATION

Using this formula to create calculated columns

Availability Status = IF('Bike Station'[Available Bikes]=0,"No Bikes",IF('Bike Station'[Available Bikes]<15,"Low Availability","Available"))

Station Category = IF('Bike Station'[Bike Stands]>=40,"Large",IF('Bike Station'[Bike Stands]>=20,"Medium","Small"))

Occupied Bikes per Station = ('Bike Station'[Bike Stands] - 'Bike Station'[Available Bike Stands])

utilization Level = IF('Bike Station'[Available Bikes]<=5,"High Usage",IF('Bike Station'[Available Bikes]<=15,"Moderate Usage","Low Usage"))

Calculated Columns

1 utilization Level = IF('Bike Station'[Available Bikes]<=5,"High Usage",IF('Bike Station'[Available Bikes]<=15,"Moderate Usage","Low Usage"))

Available Bikes	Latitude	Longitude	Station Key	Availability Status	Stand Category	Occupied Bikes per Station	utilization Level
3	37.4129235511391	-5.98890593824315	seville-1	Low Availability	Medium	3	High Usage
20	37.407071	-6.002969	seville-6	Available	Medium	20	Low Usage
4	37.407524	-6.007428	seville-7	Low Availability	Medium	5	High Usage
14	37.401499	-6.009033	seville-8	Low Availability	Medium	15	Moderate Usage
19	37.402863	-6.004107	seville-9	Available	Medium	19	Low Usage
0	37.409109	-5.990943	seville-10	No Bikes	Medium	1	High Usage
14	37.40665	-5.993109	seville-11	Low Availability	Medium	14	Moderate Usage
7	37.406927	-5.982622	seville-16	Low Availability	Medium	7	Moderate Usage
1	37.4071515555687	-5.97884017277645	seville-17	Low Availability	Medium	3	High Usage
5	37.3820754594393	-5.96812112073878	seville-18	Low Availability	Medium	5	High Usage
2	37.4033275365855	-5.98903720634574	seville-19	Low Availability	Medium	2	High Usage
4	37.4054735478923	-5.98152718123615	seville-21	Low Availability	Medium	4	High Usage
1	37.4051295489667	-5.97483115755776	seville-22	Low Availability	Medium	2	High Usage
0	37.382654	-5.97113	seville-30	No Bikes	Medium	0	High Usage
10	37.400868	-5.993751	seville-35	Low Availability	Medium	10	Moderate Usage
0	37.40062	-5.998235	seville-36	No Bikes	Medium	0	High Usage
9	37.398795	-5.993472	seville-37	Low Availability	Medium	9	Moderate Usage
3	37.3978400223044	-5.99850468687796	seville-39	Low Availability	Medium	5	High Usage
14	37.3968655090074	-5.99392721978722	seville-49	Low Availability	Medium	15	Moderate Usage
2	37.3955195005648	-6.0023212484675	seville-51	Low Availability	Medium	2	High Usage
9	37.393017	-5.986861	seville-56	Low Availability	Medium	9	Moderate Usage
12	37.3921748	-5.9910914	seville-58	Low Availability	Medium	12	Moderate Usage

Calculate Measure

3K Total Stations	57K Total Bike Stands	32K Total Available Bike...	20K Total Available Bikes	3K Open Stations Count	206 Closed Stations Count
12-11-2025... Latest Update	0.44 Occupancy %	20.03 Avg Stand per Station	336 Banking Stations	11.76 Banking %	

DATA MODELING

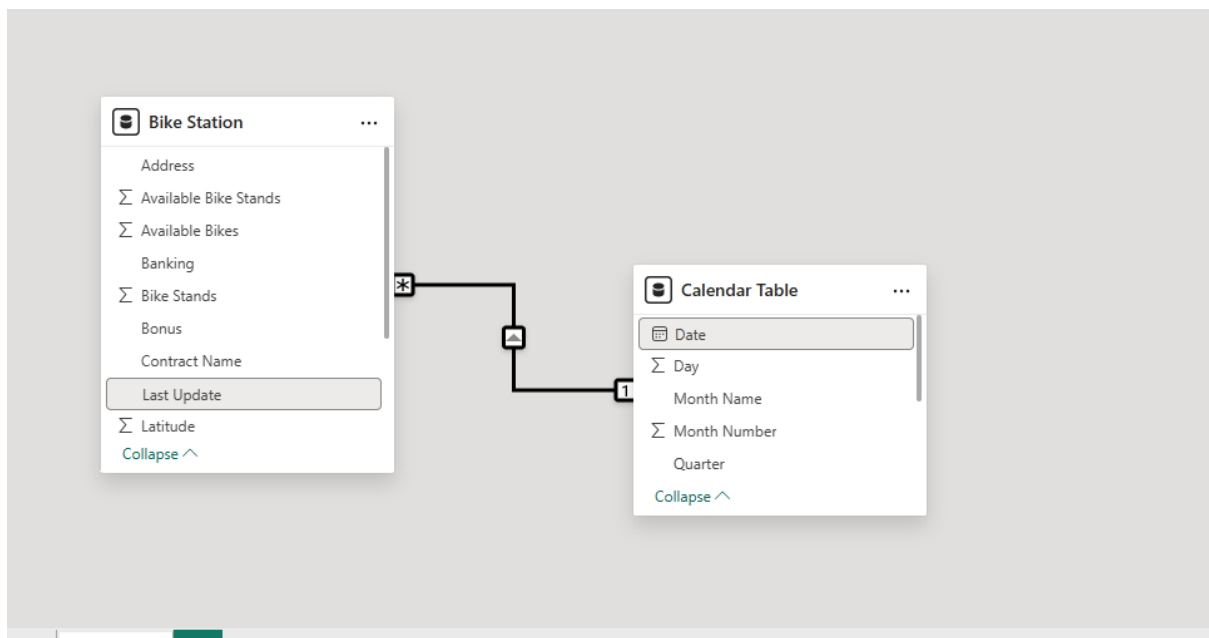
Create a Calendar Table

1 Calendar Table = ADDCOLUMNS(
2 CALENDAR(MIN('Bike Station'[Last Update]),MAX('Bike Station'[Last Update])),
3 "Year", YEAR([Date]),
4 "Month Number", MONTH([Date]),
5 "Month Name", FORMAT([Date], "MMM"),
6 "Year-Month", FORMAT([Date], "YYYY-MM"),
7 "Quarter", "Q" & FORMAT([Date], "Q"),
8 "Day", DAY([Date]),
9 "Weekday", FORMAT([Date], "DDD"),
10 "Weekday Number", WEEKDAY([Date], 2)
11)

Date	Year	Month Number	Month Name	Year-Month	Quarter	Day	Weekday	Weekday Number
01-07-2022	2022	7	Jul	2022-07	Q3	1	Fri	5
02-07-2022	2022	7	Jul	2022-07	Q3	2	Sat	6
03-07-2022	2022	7	Jul	2022-07	Q3	3	Sun	7
04-07-2022	2022	7	Jul	2022-07	Q3	4	Mon	1
05-07-2022	2022	7	Jul	2022-07	Q3	5	Tue	2
06-07-2022	2022	7	Jul	2022-07	Q3	6	Wed	3
07-07-2022	2022	7	Jul	2022-07	Q3	7	Thu	4
08-07-2022	2022	7	Jul	2022-07	Q3	8	Fri	5
09-07-2022	2022	7	Jul	2022-07	Q3	9	Sat	6
10-07-2022	2022	7	Jul	2022-07	Q3	10	Sun	7
11-07-2022	2022	7	Jul	2022-07	Q3	11	Mon	1
12-07-2022	2022	7	Jul	2022-07	Q3	12	Tue	2
13-07-2022	2022	7	Jul	2022-07	Q3	13	Wed	3
14-07-2022	2022	7	Jul	2022-07	Q3	14	Thu	4

Calendar Table (1,395 rows) Column: Date (1,395 distinct values)

Relationship between Bike Station table and Calendar Table



DASHBOARD



Bike Station Utilization Analysis

20.03

Avg Stand pe...

3K

Total Stations

57K

Total Bike Stands

20K

Total Available Bikes

12-11-202...

Latest Update

206

Closed Stations Count

Banking

False

True

Availability Status

☐

Available

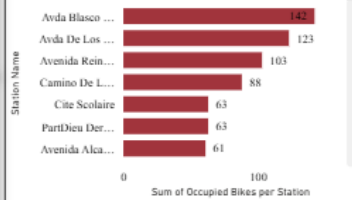
☐

Low Availability

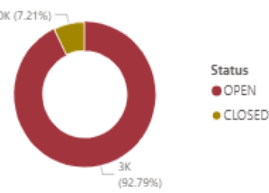
☐

No Bikes

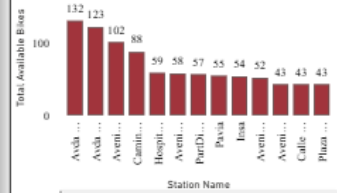
Bike Occupancy Distribution



Station Status Distribution



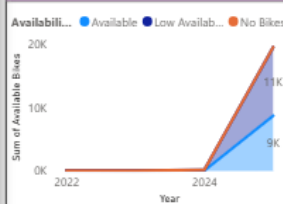
Top Stations by Available Bikes



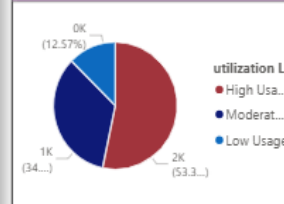
Bike Availability & Station Status Map



Annual Bike Availability Trends



Total Stations by utilization Level



Available Bikes by Stand Category

